

SmartCEM

Continuous Emissions Monitoring System

SCU Memory Map for use with GCEM4000/4100

OPS.088 – SCU Memory Map
Issue : A
Revision :
Date : 22 November 2007
Doc i/d : 0088/6
070054





Total Solutions – Total Confidence

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SCU Memory Map

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Model: SCU cem M Software ID: 507-035ACI Module: MM0100

Memory Map: Gcem Output Data

Location	Description
0100 }	Ch1 ppm
0101 }	
0102 }	Ch2 ppm
0103 }	
0104 }	Ch3 ppm
0105 }	
0106 }	Ch4 ppm
0107 }	
0108 }	Ch5 ppm
0109 }	
010A }	Ch6 ppm
010B }	
010C }	Ch7 ppm
010D }	
010E }	Temperature }
010F }	}
0110 }	Oxygen }
0111 }	}
0112 }	Pressure }
0113 }	}
0114 }	Water Vapour }
0115 }	}
0116 }	Flue temperature}
0117 }	
0118	Channel data valid
0119	Operation data valid
011A	Purging Status (01=power up,02=plant off,0C=precal,08=in cal)
011B	Calibration in progress
011C	Plant Status
011D	Head identifier (04)
011E	Reserved for SCU comms validation 00 = o.k., FF = fail
011F	

Model: SCU cem M Software ID: 507-035ACI Module: MM0120

Memory Map: Dcem 2000 data

Location	Description	
0120 }	% Opacity (*100)	
0121 }		
0122 }	Extinction (*10000)	
0123 }		
0124		
0125		
0126 }	Ch.1	}
0127 }		}
0128 }	Ch.2	}
0129 }		}
012A }	Ch.3	}
012B }		}
012C }	Ch.4	}
012D }		}
012E }	Temperature	}
012F }	Ch8	}
0130 }	Oxygen/CO2	}
0131 }	Ch6	}
0132 }	Pressure	}
0133 }	Ch7	}
0134 }	Water Vapour	}
0135 }	Ch5	}
0136		
0137		
0138		
0139		
013A	Data Valid	
013B	Calibration in progress	
013C	Plant Status	
013D	Head identifier 05= Dcem2000	
013E	Reserved for SIU comms validation 00 = o.k., FF = fail	
013F		

Analogue inputs

(Optional using 802-204A input card in Dcem for use when Gcem not present)

Normalising inputs

Model: SCU cem M Software ID: 507-035ACI Module: MM0140

Memory Map: Vcem 5000 data

Location	Description		
0140 }	Flow m/s(*10)		
0141 }			
0142			
0143			
0144			
0145			
0146			
0147			
0148 }	sample flow rate	}	flow & temperatures copied from SCU
0149 }	(*100 l/min)	}	
014A }	sample line temp.	}	
014B }	(100 degC)	}	
014C }	probe temp	}	
014D }	(100 degC)	}	
014E }	cabinet temp.	}	
014F }	(100 degC)	}	
0150	Ch1	}	Logic input data from card in SCU (7EE2 = 01)
0151	Ch2	}	
0152	Ch3	}	
0153	Ch4	}	
0154	Ch5	}	
0155	Ch6	}	
0156	Ch7	}	
0157	Ch8	}	
0158			
0159			
015A	Data Valid		
015B	Calibration in progress		
015C	Plant Status		
015D	Head identifier 06= Vcem5000		
015E	Reserved for SIU comms validation 00 = o.k., FF = fail		
015F			

Model: SCU cem M Software ID: 507-035ACI Module: MM0160

Memory Map: Icem, IOcem file

Location	Description	
0160 }	Ch1	]
0161 }		]
0162 }	Ch2	]
0163 }		]
0164 }	Ch3	]
0165 }		]
0166 }	Ch4	]
0167 }		]
0168 }	Ch5	]
0169 }		]
016A }	Ch6	]
016B }		]
016C }	Ch7	]
016D }		]
016E }	Ch8	]
016F }		]
0170	Ch1	]
0171	Ch2	]
0172	Ch3	]
0173	Ch4	]
0174	Ch5	]
0175	Ch6	]
0176	Ch7	]
0177	Ch8	]
0178	Condensed logic output data (IOcem only)	
0179		
017A		
017B		
017C		
017D	Head identifier 07= Icem, 0A= IOcem	
017E	Reserved for SIU comms validation 00 = o.k., FF = fail	
017F		

Model: SCU cem M Software ID: 507-035ACI Module: MM0180

Memory Map: vpm data

Location	Description		
0180 }	Ch1	]	
0181 }		]	
0182 }	Ch2	]	
0183 }		]	
0184 }	Ch3	]	3min (Analyser response)
0185 }		]	
0186 }	Ch4	]	
0187 }		]	
0188 }	Ch5	]	
0189 }		]	
018A }	Ch6	]	
018B }		]	
018C }	Ch7	]	
018D }		]	
018E }	Opacity	]	
018F }		]	
0190 }	Ch1	]	
0191 }		]	
0192 }	Ch2	]	
0193 }		]	
0194 }	Ch3	]	x Minute response
0195 }		]	
0196 }	Ch4	]	
0197 }		]	
0198 }	Ch5	]	
0199 }		]	
019A }	Ch6	]	
019B }		]	
019C }	Ch7	]	
019D }		]	
019E }	Opacity	]	
019F }		]	

Model: SCU cem M Software ID: 507-035ACI Module: MM01A0

Memory Map: mg/Nm3 data

Location	Description	
01A0 }	Ch1	]
01A1 }		]
01A2 }	Ch2	]
01A3 }		]
01A4 }	Ch3	]
01A5 }		]
01A6 }	Ch4	]
01A7 }		]
01A8 }	Ch5	]
01A9 }		]
01AA }	Ch6	]
01AB }		]
01AC }	Ch7	]
01AD }		]
01AE }	Dust	]
01AF }		]
01B0 }	Ch1	]
01B1 }		]
01B2 }	Ch2	]
01B3 }		]
01B4 }	Ch3	]
01B5 }		]
01B6 }	Ch4	]
01B7 }		]
01B8 }	Ch5	]
01B9 }		]
01BA }	Ch6	]
01BB }		]
01BC }	Ch7	]
01BD }		]
01BE }	Dust	]
01BF }		]

3min (Analyser response)

x Minute response

Model: SCU cem M Software ID: 507-035ACI Module: MM01C0

Memory Map: mass flow:- kg/hr, g/s, g/h or Tonnes/year data

Location	Description	
01C0 }	Ch1	]
01C1 }		]
01C2 }	Ch2	]
01C3 }		]
01C4 }	Ch3	]
01C5 }		]
01C6 }	Ch4	]
01C7 }		]
01C8 }	Ch5	]
01C9 }		]
01CA }	Ch6	]
01CB }		]
01CC }	Ch7	]
01CD }		]
01CE }	Dust	]
01CF }		]
01D0 }	Ch1	]
01D1 }		]
01D2 }	Ch2	]
01D3 }		]
01D4 }	Ch3	]
01D5 }		]
01D6 }	Ch4	]
01D7 }		]
01D8 }	Ch5	]
01D9 }		]
01DA }	Ch6	]
01DB }		]
01DC }	Ch7	]
01DD }		]
01DE }	Dust	]
01DF }		]

3min (Analyser response)

x Minute response

Model: SCU cem M Software ID: 507-035ACI Module: MM01E0

Memory Map: Other data

Location	Description	
01E0 }	Temperature degC	}
01E1 }		}
01E2 }	Oxygen/CO2 dry %(*10)	}
01E3 }		}
01E4 }	Pressure kPa	}
01E5 }		}
01E6 }	Water Vapour %(*10)	}
01E7 }		}
01E8 }	Opacity %(*100)	
01E9 }		
01EA }	Extinction (*10000)	
01EB }		
01EC }	Flow m/s(*10)	
01ED }		
01EE }	Flow Nm/s(*10)	
01EF }		
01F0 }	Flow m3/s	
01F1 }		
01F2 }	Flow Nm3/s	
01F3 }		
01F4 }	Flow Nm3/hr(*1000)	
01F5 }		
01F6 }	spare	
01F7 }	--MODBUS MAP END--	
01F8 }	Gcem all channels comprehensive data valid	
01F9 }	Dcem comprehensive data valid	
01FA }	Vcem comprehensive data valid	
01FB }	IOcem data valid	
01FC }	Gcem purge initiation flag	
01FD }	Ambient purge initiation flag	
01FE }	Back purge initiation flag	
01FF }	Reserved for maintenance flag	

actual normalisation
values

Model: SCU cem M Software ID: 507-035ACI Module: MM7E00

Memory Map: Internal EEROM

Location	Description
7E00 }	Span Gas 1 value ppm(*1) / %(*10) (Not Used)
7E01 }	
7E02 }	Span Gas 2 value ppm(*1) / %(*10) (Not Used)
7E03 }	
7E04 }	Span Gas 3 value ppm(*1) / %(*10) (Not Used)
7E05 }	
7E06 }	Span Gas 4 value ppm(*1) / %(*10) (Not Used)
7E07 }	
7E08 }	Span Gas 5 value ppm(*1) / %(*10) (Not Used)
7E09 }	
7E0A }	Span Gas 6 value ppm(*1) / %(*10) (Not Used)
7E0B }	
7E0C }	Span Gas 7 value ppm(*1) / %(*10) (Not Used)
7E0D }	
7E0E }	vpm offset (for MCERT testing only)
7E0F }	0001 = 1vpm or 0.1 %vol
7E10	
7E11	
7E12	
7E13	
7E14	
7E15	
7E16	
7E17	
7E18 }	} AWC temperature lower limit (degC *100)
7E19 }	}
7E1A }	} AWC temperature upper limit (degC *100)
7E1B }	}
7E1C	
7E1D	
7E1E	
7E1F	Reset counter (incremented every time microprocessor is reset)

Model: SCU cem M Software ID: 507-035ACI Module: MM7E20

Memory Map: Internal EEROM

Location	Description
7E20 }	Inlet Zone temperature/AWC control parameters
7E21 }	
7E22	Proportional constant
7E23	Integral constant
7E24	Differential constant
7E25	Time constant n
7E26 }	Mark/space control value
7E27 }	
7E28	Control flag (01 for auto, 00 for manual)
7E29	Inlet Zone Temperature smoothing factor
7E2A }	Zone#1 temperature
7E2B }	
7E2C	Proportional constant
7E2D	Integral constant
7E2E	Differential constant
7E2F	Time constant n
7E30 }	Mark/space control value
7E31 }	
7E32	Control flag (01 for auto, 00 for manual)
7E33	Zone#1 Temperature smoothing factor
7E34 }	Zone#2 temperature
7E35 }	
7E36	Proportional constant
7E37	Integral constant
7E38	Differential constant
7E39	Time constant n
7E3A }	Mark/space control value
7E3B }	
7E3C	Control flag (01 for auto, 00 for manual)
7E3D	Zone#2 Temperature smoothing factor
7E3E	
7E3F	

Model: SCU cem M Software ID: 507-035ACI Module: MM7E40

Memory Map: Internal EEROM

Location	Description
7E40 }	Zone#3 temperature
7E41 }	
7E42	Proportional constant
7E43	Integral constant
7E44	Differential constant
7E45	Time constant n
7E46 }	Mark/space control value
7E47 }	
7E48	Control flag (01 for auto, 00 for manual)
7E49	Zone#3 Temperature smoothing factor
7E4A }	Zone#4 temperature
7E4B }	
7E4C	Proportional constant
7E4D	Integral constant
7E4E	Differential constant
7E4F	Time constant n
7E50 }	Mark/space control value
7E51 }	
7E52	Control flag (01 for auto, 00 for manual)
7E53	Zone#4 Temperature smoothing factor
7E54 }	Sample tube temperature
7E55 }	
7E56	Proportional constant
7E57	Integral constant
7E58	Differential constant
7E59	Time constant n
7E5A }	Mark/space control value
7E5B }	
7E5C	Control flag (01 for auto, 00 for manual)
7E5D	Sample tube Temperature smoothing factor
7E5E	Sample tube temperature near % factor
7E5F	Sample tube temperature far % factor

Model: SCU cem M Software ID: 507-035ACI Module: MM7E60

Memory Map: Internal EEROM :(801-174 electro-mechanical RELAY config)

Location	Description	
7E60 }	Absolute data loc. }	
7E61 }		
7E62 }	Averaging value }	
7E63 }	Direction }	Alarm 1
7E64 }	Level }	
7E65 }		
7E66 }	Source (00=1byte) }	
7E67 }	Spare }	
7E68 }	Absolute data loc. }	
7E69 }		
7E6A }	Averaging value }	
7E6B }	Direction }	Alarm 2
7E6C }	Level }	
7E6D }		
7E6E }	Source (00=1byte) }	
7E6F }	Spare }	
7E70 }	Absolute data loc. }	
7E71 }		
7E72 }	Averaging value }	
7E73 }	Direction }	Alarm 3
7E74 }	Level }	
7E75 }		
7E76 }	Source (00=1byte) }	
7E77 }	Spare }	
7E78 }	Absolute data loc. }	
7E79 }		
7E7A }	Averaging value }	
7E7B }	Direction }	Alarm 4
7E7C }	Level }	
7E7D }		
7E7E }	Source (00=1byte) }	
7E7F }	Spare }	

Model: SCU cem M Software ID: 507-035ACI Module: MM7E80

Memory Map: Internal EEROM

Location	Description
7E80 00	Span verification available (01=yes, 00=no) (IOcem or SSR card in scu)
7E81 01 (CO)	Span 1 gas species
7E82 03 (NO)	Span 2 gas species
7E83 06 (SO2)	Span 3 gas species
7E84 08 (CH4)	Span 4 gas species
7E85	Span 5 gas species
7E86	Span 6 gas species
7E87	Span 7 gas species
7E88 32	Gas temp. No.1 % factor
7E89 32	Gas temp. No.2 % factor
7E8A 10	Gas temp. smoothing factor
7E8B 10	Probe temp. smoothing factor
7E8C 10	Cabinet temp. smoothing factor
7E8D	
7E8E 10	Back Purge Cycles
7E8F 02	Back Purge Period
7E90 F0	Power on period before Gcem cal (1unit = 1minute)
7E91 1C	Plant on period before Gcem cal (1unit = 1minute)
7E92 05	Pre/post span zero purge period (1unit = 1minute)
7E93 05	Span check purge period (1unit = 1minute)
7E94 01	Cal type flag head #1
7E95 0E	Cal type flag head #2
7E96 03	Cal type flag head #3
7E97 01	Cal type flag head #4
7E98	Hours]
7E99	Day] Calibration reference time
7E9A	Month]
7E9B	Year]
7E9C	t1]
7E9D	t2] Calibration intervals (00 inhibited, 01h = 1hour)
7E9E	t3]
7E9F	t4]

Model: SCU cem M Software ID: 507-035ACI Module: MM7EA0-1

Memory Map: Internal EEROM : (mA input card installed, 7EE1=01)

Location	Description		
7EA0 }	Input scaling factor	]	
7EA1 }		]	
7EA2 }	Lower limit	]	Extra
7EA3 }		]	input ch5
7EA4 }	Upper limit	]	parameters
7EA5 }		]	
7EA6 }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EA7 }	Smoothing constant	]	
7EA8 }	Input scaling factor	]	
7EA9 }		]	
7EAA }	Lower limit	]	Extra
7EAB }		]	input ch6
7EAC }	Upper limit	]	parameters
7EAD }		]	
7EAE }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EAF }	Smoothing constant	]	
7EB0 }	Input scaling factor	]	
7EB1 }		]	
7EB2 }	Lower limit	]	Extra
7EB3 }		]	input ch7
7EB4 }	Upper limit	]	parameters
7EB5 }		]	
7EB6 }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EB7 }	Smoothing constant	]	
7EB8 }	Input scaling factor	]	
7EB9 }		]	
7EBA }	Lower limit	]	Extra
7EBB }		]	input ch8
7EBC }	Upper limit	]	parameters
7EBD }		]	
7EBE }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EBF }	Smoothing constant	]	

Model: SCU cem M Software ID: 507-035ACI Module: MM7EA0-2

Memory Map: Internal EEROM : (mA outputs 9-12 installed, 7EE1=02)

Location	Description		
7EA0 }	DAC span	}	
7EA1 }		}	Ch.9
7EA2 }	DAC zero	}	
7EA3 }		}	
7EA4 }	DAC span	}	
7EA5 }		}	Ch.10
7EA6 }	DAC zero	}	
7EA7 }		}	
7EA8 }	DAC span	}	
7EA9 }		}	Ch.11
7EAA }	DAC zero	}	
7EAB }		}	
7EAC }	DAC span	}	
7EAD }		}	Ch.12
7EAE }	DAC zero	}	
7EAF }		}	
7EB0 }	Absolute data loc.	}	0100h=NO addr 01
7EB1 }		}	0120h=Opacity addr 02
7EB2 }	0/4mA (00=0mA 01=4mA)	}	
7EB3 }	reserved	}	
7EB4 }	Averaging value	}	
7EB5 }	Span	}	O/P 9
7EB6 }		}	
7EB7 }	Zero	}	
7EB8 }		}	
7EB9 }	spare	}	
7EBA }	Absolute data loc.	}	0100h=NO addr 01
7EBB }		}	0120h=Opacity addr 02
7EBC }	0/4mA (00=0mA 01=4mA)	}	
7EBD }	reserved	}	
7EBE }	Averaging value	}	
7EBF }	Span	}	O/P 10

Model: SCU cem M Software ID: 507-035ACI Module: MM7EC0-1

Memory Map: Internal EEROM : (mA input card installed, 7EE1=01)

Location	Description		
7EC0 }	Input scaling factor	]	
7EC1 }		]	
7EC2 }	Lower limit	]	Extra
7EC3 }		]	input ch5
7EC4 }	Upper limit	]	parameters
7EC5 }		]	
7EC6 }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EC7 }	Smoothing constant	]	
7EC8 }	Input scaling factor	]	
7EC9 }		]	
7ECA }	Lower limit	]	Extra
7ECB }		]	input ch6
7ECC }	Upper limit	]	parameters
7ECD }		]	
7ECE }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7ECF }	Smoothing constant	]	
7ED0 }	Input scaling factor	]	
7ED1 }		]	
7ED2 }	Lower limit	]	Extra
7ED3 }		]	input ch7
7ED4 }	Upper limit	]	parameters
7ED5 }		]	
7ED6 }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7ED7 }	Smoothing constant	]	
7ED8 }	Input scaling factor	]	
7ED9 }		]	
7EDA }	Lower limit	]	Extra
7EDB }		]	input ch8
7EDC }	Upper limit	]	parameters
7EDD }		]	
7EDE }	Type	]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EDF }	Smoothing constant	]	

Model: SCU cem M Software ID: 507-035ACI Module: MM7EC0-2

Memory Map: Internal EEROM : (mA outputs 9-12 installed, 7EE1=02)

Location	Description		
7EC0 }		}	O/P 10
7EC1 }	Zero	}	
7EC2 }		}	
7EC3 }	spare		
7EC4 }	Absolute data loc.	}	0100h=NO addr 01
7EC5 }		}	0120h=Opacity addr 02
7EC6 }	0/4mA (00=0mA 01=4mA)	}	
7EC7 }	reserved	}	
7EC8 }	Averaging value	}	
7EC9 }	Span	}	O/P 11
7ECA }		}	
7ECB }	Zero	}	
7ECC }		}	
7ECD }	spare		
7ECE }	Absolute data loc.	}	0100h=NO addr 01
7ECF }		}	0120h=Opacity addr 02
7ED0 }	0/4mA (00=0mA 01=4mA)	}	
7ED1 }	reserved	}	
7ED2 }	Averaging value	}	
7ED3 }	Span	}	O/P 12
7ED4 }		}	
7ED5 }	Zero	}	
7ED6 }		}	
7ED7 }			
7ED8 }			
7ED9 }			
7EDA }			
7EDB }			
7EDC }			
7EDD }			
7EDE }			
7EDF }			

Model:SCU cem M Software ID: 507-035ACI Module: MM7EE0

Memory Map: Internal EEROM

Location	Description
7EE0	Solid state relay card (00= not fitted, 01= installed scu & span, 81=active weather cover & span
7EE1	mA I/O card; 00= not used, 01= mA input card, 02= mA 9-12 output card
7EE2	Contact input card; 00= not used, 01= installed
7EE3	Span verification allowed via IOcem contact inputs, 00= none, 01= allowed
7EE4	Solid state relay mask bits 80=inverted purge 0=not inverted
7EE5	Low Sample Line flow triggers:00=nil,01=Data Invalid,02=Back Purge,04=Zero Cal
7EE6	Back purge before timed Auto cal 00= no, 01= yes
7EE7	Span checks available while plant is off (01=not, 00=yes)
7EE8	
7EE9	
7EEA	
7EEB	
7EEC }	Sample Flow 'm' factor [$\text{l/min}(*100) = \text{m}/1000 (P)^{1/2}$]
7EED }	
7EEE }	
7EEF }	Sample Flow threshold (flow < threshold => line blocked) ($\text{l/min}(*100) 450 = 4.50 \text{ l/min}$)
7EF0	Ch.1 detector locations]
7EF1	Ch.2 detector locations]
7EF2	Ch.3 detector locations]
7EF3	Ch.4 detector locations]
7EF4	Ch.5 detector locations]
7EF5	Ch.6 detector locations]
7EF6	Ch.7 detector locations]
7EF7	Ch.1 definition]
7EF8	Ch.2 definition]
7EF9	Ch.3 definition]
7EFA	Ch.4 definition]
7EFB	Ch.5 definition]
7EFC	Ch.6 definition]
7EFD	Ch.7 definition]
7EFE	Ch.7 definition]

Reserved
to be same as Gcem

Model: SCU cem M Software ID: 507-035ACI Module: MM7F00

Memory Map: Internal EEROM

Location	Description
7F00	Dust mg/m3 flag (00=mg/m3, 01=mg/Nm3)
7F01	NO/NOx flag (00=NO, 01=NOx) NOvpm, NO or NOx mg/m3, mg/Nm3 & mass emissions
7F02 }	NO2 % (*10) [used for NOx calculations when NO2 ch. not present]
7F03 }	
7F04	Plant status off (01=zero, 00=live) vpm, mg/m3, mg/Nm3 & mass emission outputs
7F05	Gcem span checks available while plant is off (01=not, 00=yes)
7F06	O2/CO2 Normalisation flag (00=O2, 01=CO2)
7F07	O2/CO2 dry/wet flag (00=DRY, 01=WET)
7F08 }	Temperature }
7F09 }	}
7F0A }	Oxygen %(*10) }
7F0B }	}
7F0C }	Pressure }
7F0D }	}
7F0E }	Water Vapour (Water Vapour dry/wet 00=DRY, 01=WET flag)
7F0F }	
7F10 }	Dust factor A }
7F11 }	}
7F12 }	Dust factor B }
7F13 }	}
7F14 }	Cross Sectional Area m2 (*100)
7F15 }	for volumetric flow
7F16	
7F17	
7F18	
7F19	
7F1A	
7F1B	
7F1C	
7F1D	
7F1E	
7F1F	Reference EEROM Location (01 = SET DEFAULT VALUES AT POWER UP)

Model: SCU cem M Software ID: 507-035ACI Module: MM7F20

Memory Map: Internal EEROM

Location	Description	
7F20 }	Divisor	]
7F21 }		]
7F22 }	Multiplier	]
7F23 }		]
7F24 }	Divisor	]
7F25 }		]
7F26 }	Multiplier	]
7F27 }		]
7F28 }	Divisor	]
7F29 }		]
7F2A }	Multiplier	]
7F2B }		]
7F2C }	Divisor	]
7F2D }		]
7F2E }	Multiplier	]
7F2F }		]
7F30 }	Divisor	]
7F31 }		]
7F32 }	Multiplier	]
7F33 }		]
7F34 }	Divisor	]
7F35 }		]
7F36 }	Multiplier	]
7F37 }		]
7F38 }	Divisor	]
7F39 }		]
7F3A }	Multiplier	]
7F3B }		]
7F3C }	Divisor	]
7F3D }		]
7F3E }	Multiplier	]
7F3F }		]

Mass flow ch1 = $\text{mg/m}^3 \times \text{vol. flow} \times \frac{\text{Multiplier}}{\text{Divisor}}$

Mass flow ch2

Mass flow ch3

Mass flow ch4

Mass flow ch5

Mass flow ch6

Mass flow ch7

Mass flow ch8

Model: SCU cem M Software ID: 507-035ACI Module: MM7F40

Memory Map: Internal EEROM :- O/P parameters

Location	Description
7F40	Averaging rate #0h
7F41	Averaging rate #1h
7F42	Averaging rate #2h
7F43	Averaging rate #3h
7F44	Averaging rate #4h
7F45	Averaging rate #5h
7F46	Averaging rate #6h
7F47	Averaging rate #7h
7F48	Averaging rate #8h
7F49	Averaging rate #9h
7F4A	Averaging rate #Ah
7F4B	Averaging rate #Bh
7F4C	Averaging rate #Ch
7F4D	Averaging rate #Dh
7F4E	Averaging rate #Eh
7F4F	Averaging rate #Fh
7F50	Averaging rate #10h
7F51	Averaging rate #11h
7F52	Averaging rate #12h
7F53	Averaging rate #13h
7F54	Averaging rate #14h
7F55	Averaging rate #15h
7F56	Averaging rate #16h
7F57	Averaging rate #17h
7F58 }	Average update counter (686 * 262.14ms = 179.828s) approx 3mins
7F59 }	
7F5A	
7F5B	
7F5C	
7F5D	
7F5E	
7F5F	

Model: SCU cem M Software ID: 507-035ACI Module: MM7F60

Memory Map: Internal EEROM :- DAC Calibration Data

Location	Description		
7F60 }	DAC span }		
7F61 }			Ch.1
7F62 }	DAC zero }		
7F63 }			
7F64 }	DAC span }		
7F65 }			Ch.2
7F66 }	DAC zero }		
7F67 }			
7F68 }	DAC span }		
7F69 }			Ch.3
7F6A }	DAC zero }		
7F6B }			
7F6C }	DAC span }		
7F6D }			Ch.4
7F6E }	DAC zero }		
7F6F }			
7F70 }	DAC span }		
7F71 }			Ch.5
7F72 }	DAC zero }		
7F73 }			
7F74 }	DAC span }		
7F75 }			Ch.6
7F76 }	DAC zero }		
7F77 }			
7F78 }	DAC span }		
7F79 }			Ch.7
7F7A }	DAC zero }		
7F7B }			
7F7C }	DAC span }		
7F7D }			Ch.8
7F7E }	DAC zero }		
7F7F }			

Model: SCU cem M Software ID: 507-035ACI Module: MM7F80

Memory Map: Internal EEROM :-

Location	Description		
7F80 }	Absolute data loc.	}	0100h=NO addr 01
7F81 }		}	0120h=Opacity addr 02
7F82 }	0/4mA (00=0mA 01=4mA)	}	
7F83 }	reserved	}	
7F84 }	Averaging value	}	
7F85 }	Span	}	O/P 1
7F86 }		}	
7F87 }	Zero	}	
7F88 }		}	
7F89 }	Absolute data loc.	}	0100h=NO addr 01
7F8A }		}	0120h=Opacity addr 02
7F8B }	Averaging value	}	
7F8C }	Direction	}	Alarm 1
7F8D }	Level	}	
7F8E }		}	
7F8F }	Source (00=1byte)	}	
7F90 }	Absolute data loc.	}	0100h=NO addr 01
7F91 }		}	0120h=Opacity addr 02
7F92 }	0/4mA (00=0mA 01=4mA)	}	
7F93 }	reserved	}	
7F94 }	Averaging value	}	
7F95 }	Span	}	O/P 2
7F96 }		}	
7F97 }	Zero	}	
7F98 }		}	
7F99 }	Absolute data loc.	}	0100h=NO addr 01
7F9A }		}	0120h=Opacity addr 02
7F9B }	Averaging value	}	
7F9C }	Direction	}	Alarm 2
7F9D }	Level	}	
7F9E }		}	
7F9F }	Source (00=1byte)	}	

Model: SCU cem M Software ID: 507-035ACI Module: MM7FA0

Memory Map: Internal EEROM :-

Location	Description		
7FA0 }	Absolute data loc.	}	0100h=NO addr 01
7FA1 }		}	0120h=Opacity addr 02
7FA2 }	0/4mA (00=0mA 01=4mA)	}	
7FA3 }	reserved	}	
7FA4 }	Averaging value	}	
7FA5 }	Span	}	O/P 3
7FA6 }		}	
7FA7 }	Zero	}	
7FA8 }		}	
7FA9 }	Absolute data loc.	}	0100h=NO addr 01
7FAA }		}	0120h=Opacity addr 02
7FAB }	Averaging value	}	
7FAC }	Direction	}	Alarm 3
7FAD }	Level	}	
7FAE }		}	
7FAF }	Source (00=1byte)	}	
7FB0 }	Absolute data loc.	}	0100h=NO addr 01
7FB1 }		}	0120h=Opacity addr 02
7FB2 }	0/4mA (00=0mA 01=4mA)	}	
7FB3 }	reserved	}	
7FB4 }	Averaging value	}	
7FB5 }	Span	}	O/P 4
7FB6 }		}	
7FB7 }	Zero	}	
7FB8 }		}	
7FB9 }	Absolute data loc.	}	0100h=NO addr 01
7FBA }		}	0120h=Opacity addr 02
7FBB }	Averaging value	}	
7FBC }	Direction	}	Alarm 4
7FBD }	Level	}	
7FBE }		}	
7FBF }	Source (00=1byte)	}	

Model: SCU cem M Software ID: 507-035ACI Module: MM7FC0

Memory Map: Internal EEROM :-

Location	Description		
7FC0 }	Absolute data loc.	}	0100h=NO addr 01
7FC1 }		}	0120h=Opacity addr 02
7FC2 }	0/4mA (00=0mA 01=4mA)	}	
7FC3 }	reserved	}	
7FC4 }	Averaging value	}	
7FC5 }	Span	}	O/P 5
7FC6 }		}	
7FC7 }	Zero	}	
7FC8 }		}	
7FC9 }	Absolute data loc.	}	0100h=NO addr 01
7FCA }		}	0120h=Opacity addr 02
7FCB }	Averaging value	}	
7FCC }	Direction	}	Alarm 5
7FCD }	Level	}	
7FCE }		}	
7FCF }	Source (00=1byte)	}	
7FD0 }	Absolute data loc.	}	0100h=NO addr 01
7FD1 }		}	0120h=Opacity addr 02
7FD2 }	0/4mA (00=0mA 01=4mA)	}	
7FD3 }	reserved	}	
7FD4 }	Averaging value	}	
7FD5 }	Span	}	O/P 6
7FD6 }		}	
7FD7 }	Zero	}	
7FD8 }		}	
7FD9 }	Absolute data loc.	}	0100h=NO addr 01
7FDA }		}	0120h=Opacity addr 02
7FDB }	Averaging value	}	
7FDC }	Direction	}	Alarm 6
7FDD }	Level	}	
7FDE }		}	
7FDF }	Source (00=1byte)	}	

Model: SCU cem M Software ID: 507-035ACI Module: MM7FE0

Memory Map: Internal EEROM :-

Location	Description		
7FE0 }	Absolute data loc.	}	0100h=NO addr 01
7FE1 }		}	0120h=Opacity addr 02
7FE2 }	0/4mA (00=0mA 01=4mA)	}	
7FE3 }	reserved	}	
7FE4 }	Averaging value	}	
7FE5 }	Span	}	O/P 7
7FE6 }		}	
7FE7 }	Zero	}	
7FE8 }		}	
7FE9 }	Absolute data loc.	}	0100h=NO addr 01
7FEA }		}	0120h=Opacity addr 02
7FEB }	Averaging value	}	
7FEC }	Direction	}	Alarm 7
7FED }	Level	}	
7FEE }		}	
7FEF }	Source (00=1byte)	}	
7FF0 }	Absolute data loc.	}	0100h=NO addr 01
7FF1 }		}	0120h=Opacity addr 02
7FF2 }	0/4mA (00=0mA 01=4mA)	}	
7FF3 }	reserved	}	
7FF4 }	Averaging value	}	
7FF5 }	Span	}	O/P 8
7FF6 }		}	
7FF7 }	Zero	}	
7FF8 }		}	
7FF9 }	Absolute data loc.	}	0100h=NO addr 01
7FFA }		}	0120h=Opacity addr 02
7FFB }	Averaging value	}	
7FFC }	Direction	}	Alarm 8
7FFD }	Level	}	
7FFE }		}	
7FFF }	Source (00=1byte)	}	

SLAVE

Model: SCU cem S Software ID: 507-034A Module: MM7E00

Memory Map: Internal EEROM

Location			Description
7E00			No. of heads connected to SCU (01 to 04)
7E01	01		Normalisation source addr if GCEM not present in system
7E02	FF		Contact I/P FF=not used, 00=active O/C, 01=active S/C
7E03	FF		Contact I/P Selector: 01=Plant Status, 02= Gcem Air Press. Switch
7E04			
7E05			
7E06			
7E07			
7E08	01	}	Head #1 output file location GCEM
7E09	E0	}	
7E0A	01	}	Head #2 output file location DCEM
7E0B	E0	}	
7E0C	01	}	Head #3 output file location VCEM
7E0D	E0	}	
7E0E	01	}	Head #4 output file location ICEM
7E0F	E0	}	
7E10			
7E11			
7E12			
7E13			
7E14			
7E15			
7E16			
7E17			
7E18			
7E19			
7E1A			
7E1B			
7E1C			
7E1D			
7E1E			
7E1F			Reference EEROM Location